1. What is JavaScript?

JavaScript is a high-level, interpreted programming language used to create dynamic and interactive web content.

2. What are the data types in JavaScript?

Primitive: String, Number, Boolean, Null, Undefined, Symbol, BigInt

Non-Primitive: Object, Array, Function

3. What is the difference between var, let, and const?

- var: Function-scoped, can be redeclared.

- let: Block-scoped, cannot be redeclared.

- const: Block-scoped, cannot be redeclared or reassigned.

4. What is a closure?

A closure is a function that remembers variables from its outer scope even after the outer function has finished executing.

5. What is hoisting?

Hoisting is JavaScript's default behavior of moving declarations to the top of the current scope before code execution.

6. What is the difference between == and ===?

== compares values with type coercion.

=== compares both value and type (strict equality).

7. What is an arrow function?

A concise way to write functions using `=>`. It does not have its own `this` context.

Example: const add = (a, b) => a + b;

8. What is a Promise?

A Promise is an object representing the eventual completion or failure of an asynchronous operation.

9. How do you create a Promise?

```
const promise = new Promise((resolve, reject) => {
  // async code
  resolve("Success");
});
```

10. What is async/await?

Syntax to handle Promises more cleanly.

Example:

```
async function fetchData() {
  const data = await fetch(url);
}
```

11. What is the DOM?

DOM (Document Object Model) is a programming interface for HTML and XML documents. It represents the page so programs can change the document structure, style, and content.

Notes: Try to explain by drawing a tree. Refer any standard website

12. What is event bubbling?

Event bubbling is the process where an event starts from the deepest target element and propagates up to the parent elements.

13. What does preventDefault() do?

It cancels the default action of an event (e.g., stopping a form from submitting).

14. What is an object in JavaScript?

An object is a collection of key-value pairs. Example:

```
const person = { name: "John", age: 30 };
```

15. How do you create a class in JavaScript?

```
class Person {
  constructor(name) {
    this.name = name;
  }
  greet() {
    console.log(`Hello, ${this.name}`);
  }
}
```

16. What is inheritance in JavaScript?

Inheritance allows one class to inherit properties and methods from another.

Example:

```
class Student extends Person {
  study() {
    console.log(`${this.name} is studying`);
```

```
  }
}
```

17. What is the use of `this` keyword?

`this` refers to the object from which the function was called.

18. What is the callback function?

A callback is a function passed into another function as an argument and executed later.

19. What is the difference between null and undefined?

- null: intentional absence of value.
- undefined: variable declared but not assigned a value.

20. How do you handle errors in JavaScript?

Using try…catch blocks:

```
try {
  // code
} catch (error) {
  console.error(error);
}
```